Session 3 Lab - Using OLS: Interpretation, Hypothesis Testing & Confidence Intervals

This lab exercise uses the `state.x77` dataset, which contains variables related to demographics, education, and politics for the 50 U.S. states in 1977. You'll practice the key concepts from today's session: interpretation, hypothesis testing, and confidence intervals.

Setup

```
# Load the data
data(state)
# The data is in state.x77 - let's make it easier to work with
states <- data.frame(state.x77)
head(states)
```

Part 1: Data Exploration

1.1 Explore the dataset:

- How many observations and variables are there?
- What are the variable names?
- Get summary statistics for `Life.Exp` and `Income`
- What are the units for each variable? (Hint: use `?state.x77`)

1.2 Create a scatter plot of Life Expectancy vs Income:

- What relationship do you observe?
- Are there any obvious outliers?

Part 2: Simple Linear Regression

2.1 Fit a simple linear regression model with `Life.Exp` as the dependent variable and `Income` as the independent variable.

2.2 Interpret the coefficients:

- What does the intercept mean? Is this interpretation meaningful?
- What does the slope mean in practical terms?

- If income increases by $1000, what happens to life expectancy?

2.3 Examine the model output:

- What is the R-squared? What does this tell you?
- How many degrees of freedom do you have?

Part 3: Manual Hypothesis Testing

3.1 Test whether income significantly affects life expectancy:

- State your null and alternative hypotheses
- Extract the slope estimate and its standard error from your model
- Calculate the t-statistic manually
- Find the critical value for $\alpha = 0.05$
- Calculate the p-value manually using `pt()`
- Make your decision: reject or fail to reject the null?

3.2 Verify your manual calculations match the model output.

Part 4: Confidence Intervals

4.1 Calculate a 95% confidence interval for the slope coefficient:

- Do this manually using the critical value and standard error
- Verify using `confint()`
- Interpret this confidence interval correctly (remember our discussion!)

4.2 Calculate confidence intervals for different confidence levels (90%, 95%, 99%):

- How do the widths compare?
- Which would you prefer and why?

4.3 Does your confidence interval include 0? What does this tell you about the hypothesis test?

Part 5: Predictions and Residuals

5.1 Make predictions:

- Predict life expectancy for a state with income of $4000
- Predict life expectancy for a state with income of $6000

- What's the difference between these predictions? Does this make sense given your slope?

5.2 Examine specific cases:

- Find the state with the highest income and lowest income
- Calculate the predicted life expectancy for each
- Calculate the residuals for each
- Which state is performing better/worse than predicted?

5.3 Calculate and examine all residuals:

- What's the mean of the residuals? (What should it be?)
- Which states have the largest positive and negative residuals?
- What might explain these large residuals?

Part 6: Understanding Sampling Variability

6.1 Bootstrap simulation (This is advanced, but helps illustrate the sampling variability):

- Create a simple bootstrap simulation by sampling your data with replacement 100 times
- Fit the regression model to each bootstrap sample
- Plot the distribution of slope estimates
- Calculate the standard deviation of these estimates
- How does this compare to the standard error from your original model?

Part 7: Effect Sizes and Practical Significance

7.1 Assess practical significance:

- Is the effect size large enough to matter in practice?
- What's the effect of moving from the 25th to 75th percentile of income?
- Compare states at different income levels - is the predicted difference meaningful?

7.2 Put your results in context:

- How does the income effect compare to the overall variation in life expectancy?
- What factors might explain the remaining variation?